

microcomputers work. The main difference is one of physical scale. More and more transistors are being squeezed into silicon chips with each passing year and ICs are evolving with ever-increasing storage capacity and processing power. We are still not close to pushing the ultimate limit to the number of transistors per unit volume, which is imposed by the atomic structure of matter.

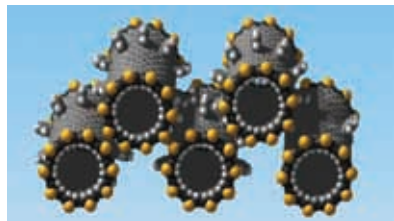
Although there has been an exponential increase in the density of transistors on integrated-circuit computer chips over the past 40 years, there has been no fundamental change in the operating principles of the transistor. They still operate using small electric fields imposed by tiny charged metal plates to control the mass action of many millions of electrons. Electronic nanocomputers are expected to continue representing information in the storage and movement of electrons, but they will be created through microscopic circuits using nanolithography.

Nanolithography is a term used to describe the branch of nanotechnology concerned with the study and application of a number of techniques for creating nanoscale structures. It is used during the fabrication of leading-edge semiconductor integrated circuits (nanocircuitry) or nanoelectromechanical systems (NEMS). Dip-pen nanolithography can achieve very small sizes, but cannot currently go below 40nm. However new nanolithography technologies are constantly being researched and developed, leading to smaller and smaller possible sizes.

## 1.2. Mechanical Nanocomputers

Pioneers as Eric Drexler proposed as far back as the mid-1980's that nanoscale mechanical computers could be built via molecular manufacturing through a process of mechanical positioning of atoms or molecular building blocks one atom or molecule at a time, a process known as "mechanosynthesis."

His proposal for mechanical nanocomputer, took inspiration from Charles Babbage's analytical engines of the 19th century. However mechanical nanocomputer technology has sparked controversy and some researchers even



Mechanosynthesis

consider it unworkable. All the problems inherent in Babbage's apparatus, according to the naysayers, are magnified a million fold in a mechanical nanocomputer. Nevertheless, some futurists are optimistic about the technology, and have even proposed the evolution of nanorobots that could operate, or be controlled by, mechanical nanocomputers.